



ETHICS OF RESEARCH IN THE CRITICALLY ILL

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ETHICS OF MEDICAL RESEARCH – A BRIEF HISTORY

Pre-WWII:

- Wild west

Post-WWII:

- Nuremberg Code
- Helsinki Declaration
 - Right to self-determination
 - Right to informed decisions
 - Institutional review boards
 - Participants welfare takes precedence over societal and scientific interests
- Ethical universalism
- Placebo unethical when proven intervention exists

ETHICS OF MEDICAL RESEARCH – CURRENTLY

Good Clinical Practice:

- Standard in Australia, US, EU
- Less principles based, more procedural

NHMRC National Statement on Ethical Conduct in Human Research:

- Australia only
- Principles:
 - Merit
 - Justice
 - Beneficence
 - Respect

VULNERABLE POPULATIONS

- Children
- Mental health populations
- Institutionalised persons
- Persons in an exploitable hierarchy
 - Students
 - Hospital and laboratory personnel
 - Military personnel
- Incurable disease
- Refugees and internally displaced persons
- Ethnic minorities
- Critically ill?

ISSUES IN THE CRITICALLY ILL

- **Consent**
 - Informed
 - Capacity
 - Voluntary
- Surrogate decision makers
- Deferred consent
- Partial consent
- Representative consent
- Waiver of consent
- Trial designs

INFORMED CONSENT

- **Informed and Specific**

- Study's aims
- Methods
- Expected benefits
- Potential harms

- **Capacity**

- Permanent incapacity
- Temporary incapacity
- Fluctuating incapacity

- **Voluntary**

- Coercion

SURROGATE DECISION MAKERS

- Surrogate decision making permitted
- Research more complicated than clinical care
- Availability of surrogate
- Capacity of surrogate
- Process that surrogate uses
 - Substituted judgement
 - Best interests

DEFERRED CONSENT

- Enrolment occurs, then consent occurs when patient regains capacity
 - Interim surrogate consent
 - Less selection bias
- May also include deferred surrogate consent

PARTIAL CONSENT

- May be used if comparison to standard of care taken
- Randomisation prior to consent
- Consent only sought in the intervention arm
- Increased power

REPRESENTATIVE CONSENT

- Enrolment after two independent physicians consent on the patients behalf

WAIVER OF CONSENT

- Permissible if research 'carries no more than low risk'
- Balance
 - Risk of participation
 - Violation of patient autonomy
 - Lost societal benefit
 - Lost opportunity for participation
- Appropriate for most retrospective designs
- Used in some other designs

WAIVER OF CONSENT - RETROSPECTIVE

- No more than low risk
- Benefit exceeds risk of harm of not seeking consent
- Consent is impractical
- No likely reason that participants would have refused
 - 93% of Australians agree that use of PHI for research is acceptable
- Privacy is protected
- Confidentiality is protected

TRIAL DESIGNS

Principles:

- Limit the intervention
- Restrict the population
- Change the population to a less vulnerable one
- Graded development

EXAMPLE - SPLIT

- Saline vs. PL-148 for AKI
- 4 ICUs randomised to masked fluid for alternating 7-week blocks
- “Standard practice” in each unit changed
- “Opt-out” consent
 - Information given
 - Patients could opt-out of use of data
- All (2278) eligible patients enrolled
- 16 opted out

Original Investigation | CARING FOR THE CRITICALLY ILL PATIENT

Effect of a Buffered Crystalloid Solution vs Saline on Acute Kidney Injury Among Patients in the Intensive Care Unit The SPLIT Randomized Clinical Trial

Paul Young, FCICM; Michael Bailey, PhD; Richard Beasley, DSc; Seton Henderson, FCICM; Diane Mackle, MN; Colin McArthur, FCICM; Shay McGuinness, FANZCA; Jan Mehrrens, RN; John Myburgh, PhD; Alex Psirides, FCICM; Sumeet Reddy, MBChB; Rinaldo Bellomo, FCICM; for the SPLIT Investigators and the ANZICS CTG

EXAMPLE – NIEMANN ET AL

- Hypothermia vs. normothermia of DBD renal transplant donors for graft function
- Donors (after death determined) cooled to 34-35 degrees
- Determined to be non-human subject research
- Consent not required from donor families – not human subject
- Consent not taken from recipient – minimal risk

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Therapeutic Hypothermia in Deceased Organ Donors and Kidney-Graft Function

Claus U. Niemann, M.D., John Feiner, M.D., Sharon Swain, M.S.N., R.N., Scott Bunting, R.R.T.,
Melissa Friedman, M.S.N., R.N., Megan Crutchfield, M.P.H., Kristine Broglio, M.S., Ryutaro Hirose, M.D.,
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EXAMPLE – PARAMEDIC2

- Adrenaline vs. placebo for 30-day survival in OOHCA
- Deferred consent from patient or surrogate
- Pre-trial refusal of consent
 - Medic-alert bracelet
- 8016 patients randomised
- 6 withdrew consent

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A Randomized Trial of Epinephrine in Out-of-Hospital Cardiac Arrest

G.D. Perkins, C. Ji, C.D. Deakin, T. Quinn, J.P. Nolan, C. Scamperin, S. Regan, J. Long, A. Slowther, H. Pocock, J.J.M. Black, F. Moore, R.T. Fothergill, N. Rees, L. O'Shea, M. Docherty, I. Gunson, K. Han, K. Charlton, J. Finn, S. Petrou, N. Stallard, S. Gates, and R. Lall, for the PARAMEDIC2 Collaborators*